RETAILPULSE

From retail transactions to clearer decisions

An end-to-end analytics and machine learning portfolio

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Sales performance / demand forecasting / customer churn / inventory

01 Project overview

Vision. Give retail teams one place to understand sales, plan demand, identify at-risk customers, and turn forecasts into inventory actions.

Objectives. Clean and explore sales, create daily and customer-level modeling datasets, train predictive models, expose results in a multipage dashboard, and retain reusable model artifacts.

Target users. Merchandising and inventory planners; CRM and retention teams; business analysts; data scientists validating model outputs.

Business value. Faster access to sales and customer patterns, a common forecast view, explainable churn signals, and reorder recommendations. Financial impact has not been measured in the repository.

Project footprint

410.8K cleaned rows | 10 dataset fields | 6 dashboard pages | 2 core model tasks

Non-functional goals (not measured)

Goal	Proposed target	How to verify
Data quality	Reject malformed dates, missing required fields, and nonnumeric revenue.	Upload validation and quality checks.
Interactive latency	Page load under 3 seconds for a typical prepared dataset.	Measure cold and warm loads.
Retraining	LSTM refresh within 5 minutes for a few thousand daily points on CPU.	Record training duration and data size.
Reliability	Failed training must not replace active model artifacts.	Failure-path and rollback checks.

02 Key features and acceptance criteria

ID	Feature	Description	Acceptance criteria
F01	CSV upload	Sidebar upload validates and activates a retail CSV for the session.	Valid schema accepted; errors shown; cleaned project dataset remains default.
F02	Sales overview	Revenue, orders, customers, products, daily trend and date coverage.	Metrics and chart follow the active dataset.
F03	Customer analytics	RFM scoring and purchase behavior segments.	Customer, invoice, date and revenue fields support a customer summary.
F04	Revenue forecast	LSTM horizon, what-if scenario and CSV download.	Forecast history uses active data; output is downloadable.
F05	LSTM retraining	Explicit user-triggered training on daily revenue.	At least 38 daily observations; update artifacts only after successful training.
F06	Churn prediction	XGBoost scoring with saved feature order and SHAP artifacts.	Requires customer features and a defensible label; generic uploads alone are insufficient.
F07	Inventory planning	Demand, safety stock and reorder thresholds.	Item-level recommendations available for planner review.
F08	Evaluation	Actual-vs-predicted metrics and model comparison.	Show task, date window and sample size with metrics.
F09	Drift / retraining ops	Evidently report and Airflow retraining prototype.	Produce saved reports; production scheduling remains future work.

03 Technology stack

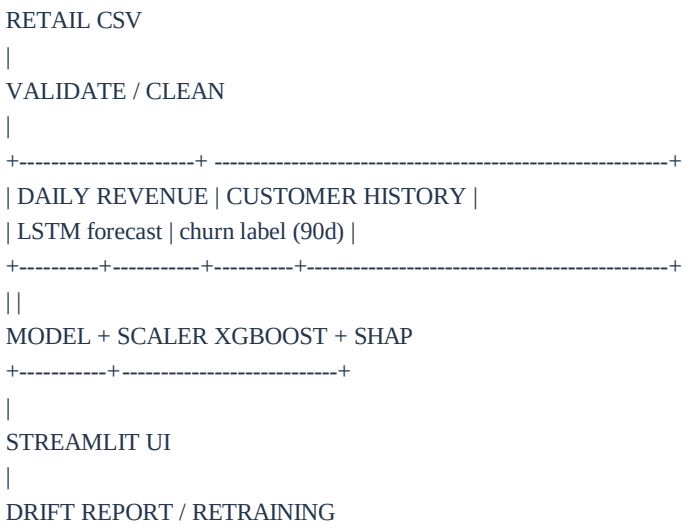
Category	Technology	Rationale / alternatives
App	Python + Streamlit	Fast multipage analytics UI. Alternative: API plus React for a larger service.
Data	Pandas, NumPy, CSV	Tabular cleaning and aggregation. Alternatives: DuckDB/Parquet or a warehouse.
Charts	Plotly, Matplotlib, Seaborn	Interactive dashboards and notebook EDA. Alternative: Altair or Power BI.
Forecasting	PyTorch, Lightning, scikit-learn; Prophet experiments	LSTM sequence model, scaling and classical time series.
Classification	XGBoost, scikit-learn, SHAP	Gradient boosted churn model with interpretation. Alternatives: LightGBM, CatBoost, calibrated logistic regression.
Tuning / drift	Optuna, Evidently	Hyperparameter search and drift reporting.
Orchestration	Apache Airflow prototype	Notebook-defined retraining DAG. Alternative: managed orchestration.
Artifacts	joblib, PyTorch state_dict, CSV/JSON	Portable local artifacts. Alternative: MLflow registry and object storage.

04 Architecture and data flow

Retail CSV -> validate and clean -> model-specific paths -> saved outputs -> Streamlit analytics. The branches use different modeling units and targets.

Data / stage	Model preparation	Output / use
Transaction CSV / cleaned retail	Type checks, deduplication, revenue and calendar features	Shared dashboard dataset
Revenue path	Aggregate Revenue by day; lag and rolling features for forecasting	LSTM model + scaler -> revenue forecasts
Churn path	Customer history features; target = no purchase in following 90-day window	XGBoost + feature order + SHAP explanations
Demand / inventory path	Product-level daily demand, forecast and safety stock calculations	Forecast CSV + reorder recommendations
MLOps path	Evidently drift report; Airflow retraining prototype	Offline report and model evaluation artifacts

HIGH-LEVEL FLOW



The churn target comes from a future outcome window; a transaction upload without target labels can support revenue retraining but does not automatically qualify for churn retraining.

05 Execution timeline

Suggested four-week delivery plan. The repository has no dated issue history, so this is a practical reconstruction and not a claim about the original calendar.

Week	Workstream	Deliverable	Checkpoint
1	Data audit + EDA	Clean extract; sales, seasonality, customer and cancellation analysis.	Agree definitions and quality rules.
2	Forecasting	Daily series; Prophet/LSTM experiments; holdout evaluation.	Review simple baseline and errors by date.
2	Churn	Historical features; 90-day churn label; XGBoost and SHAP.	Check leakage, balance and thresholds.
3	Inventory + tuning	Demand forecast, reorder logic and Optuna comparison.	Review with inventory planners.
3	Dashboard	Six pages, upload, shared active data and downloadable forecasts.	Walk through upload end-to-end.
4	MLOps + hardening	Drift report, retraining prototype, documentation and deployment plan.	Define promotion, rollback and ownership.

Milestones

M1 - data and target definitions approved. M2 - model results reviewed against baselines. M3 - dashboard uses one active dataset source. M4 - retraining and drift outputs are repeatable, with rollback defined before production.

06 Technical highlights and measured results

Revenue / demand features: daily aggregation, lags (1, 7, 14 days), rolling mean and standard deviation, and calendar features. **Churn features:** recency, invoice frequency, monetary spend, quantity, unique products, country and average order value; target is no purchase in the following 90-day window.

Saved evaluation outputs

Model / task	Metric	Recorded result	Context
XGBoost churn	ROC-AUC	0.710	Saved held-out evaluation output.
XGBoost churn	F1	0.597	Churn class; threshold-dependent.
XGBoost churn	Precision / recall	0.588 / 0.606	Review retention costs before use.
LSTM revenue	MAE / RMSE	23,519 / 24,492	Recomputed from a 7-row evaluation CSV.
LSTM revenue	MAPE / WAPE	50.7% / 60.4%	High errors; tiny sample includes a zero-actual day.
Demand RF	MAE / RMSE	132.5/222.6 -> 124.1/219.9	Saved baseline to tuned comparison.

Interpretation

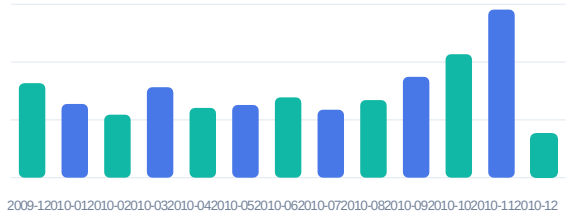
The seven-date LSTM file is too small for a broad performance claim and its errors are high. Extend the time-aware holdout and compare against seasonal-naive baselines. The tuned Random Forest improves the saved comparison slightly; that is not evidence of production lift.

07 Project visuals | 1 of 2

Charts below are report visualizations of bundled project CSV outputs, not screenshots from a live deployment.

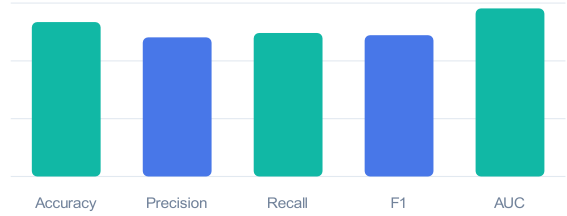
Monthly revenue profile

Aggregated from daily_revenue.csv, 2009-12 to 2010-12.



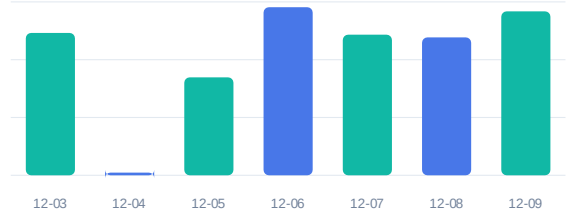
Churn model metrics

Saved evaluation CSV; values on 0-1 scale.



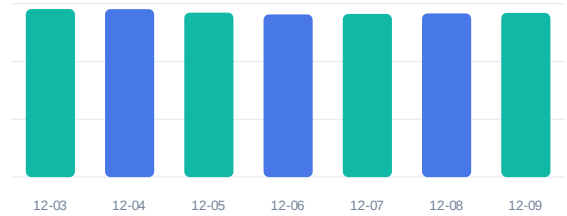
LSTM actual revenue sample

7 evaluation days; bars show actual values.



LSTM predicted revenue sample

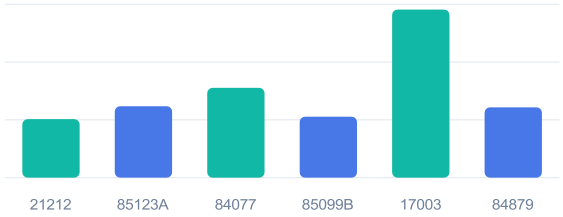
Predictions; MAE 23,519, MAPE 50.7%.



08 Project visuals | 2 of 2

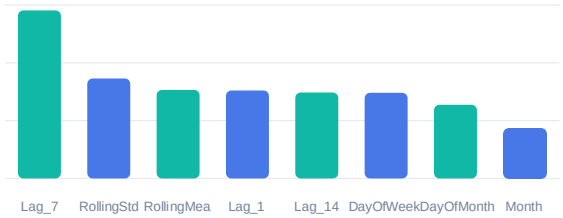
Top-item reorder point

Saved top20_inventory_recommendations.csv.

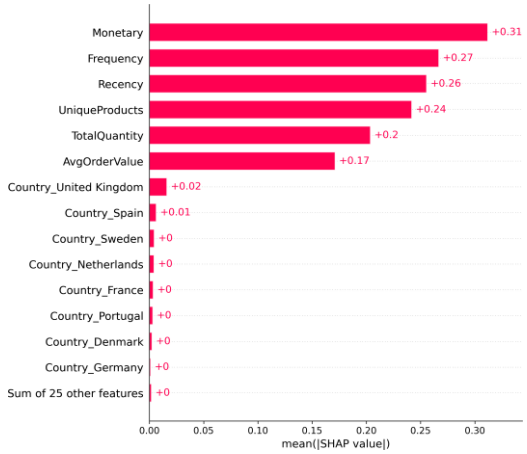


Optuna feature importance

Importance is model-specific, not causal.

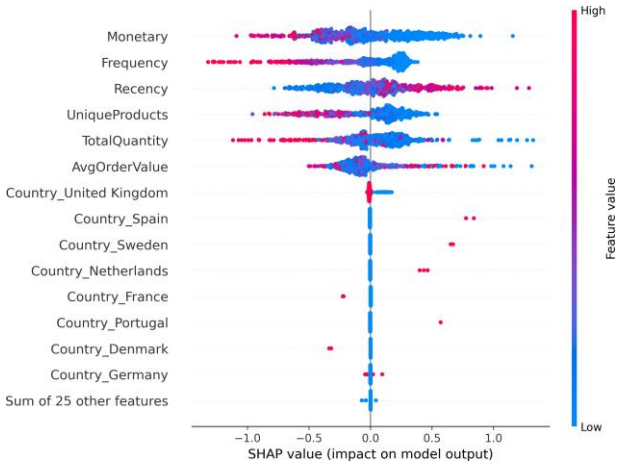


SHAP global feature importance



Saved XGBoost churn model explanation.

SHAP feature impact distribution



Dots show feature-value direction and impact; confirm the artifact matches the model version being discussed.

09 Deployment and operations

Platform. Local multipage Streamlit application. No public hosted URL or cloud deployment configuration is present.

Upload and refresh. Sidebar CSV validation activates data for the session. On Demand Forecasting, a user can explicitly retrain the LSTM from daily revenue, save a new model/scaler, and use the artifacts for later forecasts. Churn retraining needs valid labels and customer features.

Monitoring. A saved Evidently drift HTML report exists; it is offline, with no live alert channel evidenced. The Airflow notebook describes a weekly retraining DAG prototype for a demand model.

CI/CD and registry. No automated build/deploy pipeline or model registry appears in the repo. Recommended: version model/scaler together, record dataset hash and training metadata, add rollback, automate smoke checks, and deploy through controlled CI/CD.

Area	Current evidence	Next production step
Deployment	Local Streamlit and local artifacts.	Choose managed host; configure secrets, access control and persistent storage.
Retraining	Dashboard LSTM action; Airflow notebook prototype.	Run history, data fingerprint, validation gate and rollback.
Monitoring	Offline Evidently HTML report.	Schedule drift checks and alert an owner.
Registry	Model files in project folders.	MLflow or equivalent for version and lineage.
CI/CD	No workflow files found.	Add lint, checks, artifact validation and deployment approval.

No cloud vendor, public demo, CI/CD platform or live monitoring endpoint is claimed in this brief.

10 Reflection and roadmap

Key learning. Retail prediction depends on target definition as much as algorithm choice. Forecasting needs time-aware validation; churn needs customer-level features and an outcome window that prevents leakage. A transaction CSV can support revenue retraining, but cannot support churn retraining without labels or a valid derived outcome.

Practices represented. Time-aware data preparation, future-window churn labeling, stratified classification split, saved feature order, SHAP explanations, model comparison and tuning, drift reporting, model/scaler artifacts, and explicit model refresh.

Prioritized roadmap

Priority	Idea	Outcome
1	Expand LSTM holdout; add seasonal-naive baselines, MASE/WAPE by month and model-vs-baseline gates.	Trustworthy forecast evidence.
2	Version model/scaler bundles with data hash, parameters, date and rollback.	Reproducible refreshes.
3	Add churn calibration, time-based validation, threshold selection by retention economics and label freshness.	Safer retention prioritization.
4	Schedule drift reporting with thresholds and an accountable owner.	Actionable drift response.
5	Add MLflow or equivalent, CI/CD, and a hosted demo.	Governed deployment lifecycle.
6	Validate inventory recommendations against lead time, stock position and planner feedback.	Useful reorder decisions.

RetailPulse combines a useful analytics surface with model-specific data paths and interpretation. The next stage is stronger evaluation and operational governance before predictions drive automated decisions.

Project sources reviewed: Streamlit source, saved evaluation CSVs, model artifacts, data outputs, SHAP images, Evidently report and analysis notebooks.